

UMBL Research Topic List (English Edition)

The following topics are suitable for undergraduate theses, course papers, or independent research projects. All topics are aligned with UMBL's identity as a pictographic language-design development project.

1. Systematic Analysis of UMBL Character-Formation Principles

Study UMBL's rules for constructing characters and combining components.

2. Structure and Classification of Pictographic Morphemes

Explore how pictographic morphemes can be defined, categorized, and organized.

3. Combination Logic of Pictographic Words in UMBL

Analyze how components form "words" and evaluate semantic transparency.

4. Feasibility Study of Preliminary Grammar Prototypes

Attempt to construct simple grammar structures and test comprehensibility.

5. Structural Comparison Between UMBL and Chinese Character Components

Compare formation logic, semantic transparency, and compositionality.

6. Comparison of UMBL with Esperanto, Lojban, and Other Conlangs

Analyze differences in design philosophy, transparency, and learnability.

7. Study of Evolution Rules for UMBL

Investigate how to design maintainable and extensible evolution mechanisms.

8. Learning Curve Experiments for a Pictographic Language

Evaluate learning speed and difficulty through user studies.

9. Cognitive Differences Between Pictographic and Linear Text Expressions

Study differences in comprehension speed, memory load, and processing.

10. Semantic Coverage Analysis of the UMBL Component Repository

Analyze the semantic range, distribution, and gaps in the component set.

11. Evaluation of Visual Consistency and Readability in UMBL

Assess visual coherence and readability from design and cognitive perspectives.

12. Experimental Construction of Pictographic Sentences Using UMBL

Build simple sentences and test cross-cultural comprehensibility.

13. Encoding Schemes and Data Structures for UMBL

Explore computer-readable representations of pictographic language.

14. Automated Composition Algorithms for Pictographic Expressions

Design algorithms to automatically combine components into expressions.

15. Prototype Design of an Editor or Renderer for UMBL

Develop a simple editor or rendering tool for UMBL expressions.

16. Semantic Transparency Comparison Between UMBL and ISO Pictograms

Compare their effectiveness in public information communication.

17. Potential Role of UMBL in Cross-Cultural Understanding

Investigate whether pictographic language reduces cross-cultural barriers.

18. Maintainability of UMBL Character-Formation Rules

Analyze consistency and scalability of formation rules during expansion.

19. Designer Experience Study on Constructing Pictographic Units

Interview or test designers to understand challenges in constructing units.

20. User Testing of Semantic Transparency in UMBL

Evaluate intuitiveness and misinterpretation rates through experiments.

These topics can be further refined based on student interests, course requirements, or methodological preferences.

UMBL's openness and experimental nature provide rich opportunities across linguistics, cognitive science, information design, and computer science.